The document and process conversion measures necessary to comply with this revision shall be completed by 20 June 1995

INCH-POUND

MIL-S-19500/526C
20 March 1995
SUPERSEDING
MIL-S-19500/526B(USAF)
27 February 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N3879 JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, power transistors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-66).

1.3 Maximum ratings. (T_A = +25°C, unless otherwise specified).

P_T 1/ T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$
W	V dc	V dc	V dc	A dc	A dc	°C/W	°C/W
35	120	75	7	5	7	-65 to +200	5

1/ Derate linearly 200 mW/°C for T_C > +25°C.

1.4 Primary electrical characteristics.

	h_{FE2} 1/	h_{FE2} 1/	$V_{BE(SAT)1}$	$V_{CE(SAT)1}$	C_{obo}	$\|h_{fe}\|$	Switching (see table I and figure 2 herein)	
	V_{CE} = 5.0 V dc I_C = 0.5 A dc	V_{CE} = 5.0 V dc I_C = 4.0 A dc	I_C = 4.0 V dc I_B = 0.4 A dc	I_C = 4.0 V dc I_B = 0.4 A dc	V_{CB} = 10 V dc I_E = 0 0.1 MHz ≤ f ≤ 1 MHz	V_{CE} = 10 V dc I_C = 500 mA dc f = 10 MHz	t_{on}	t_{off}
			V dc	V dc	pF		µs	µs
Min	40	20				4		
Max		80	2.0	1.2	175	20	0.44	1.2

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

3.3.1 Lead material and finish. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750 and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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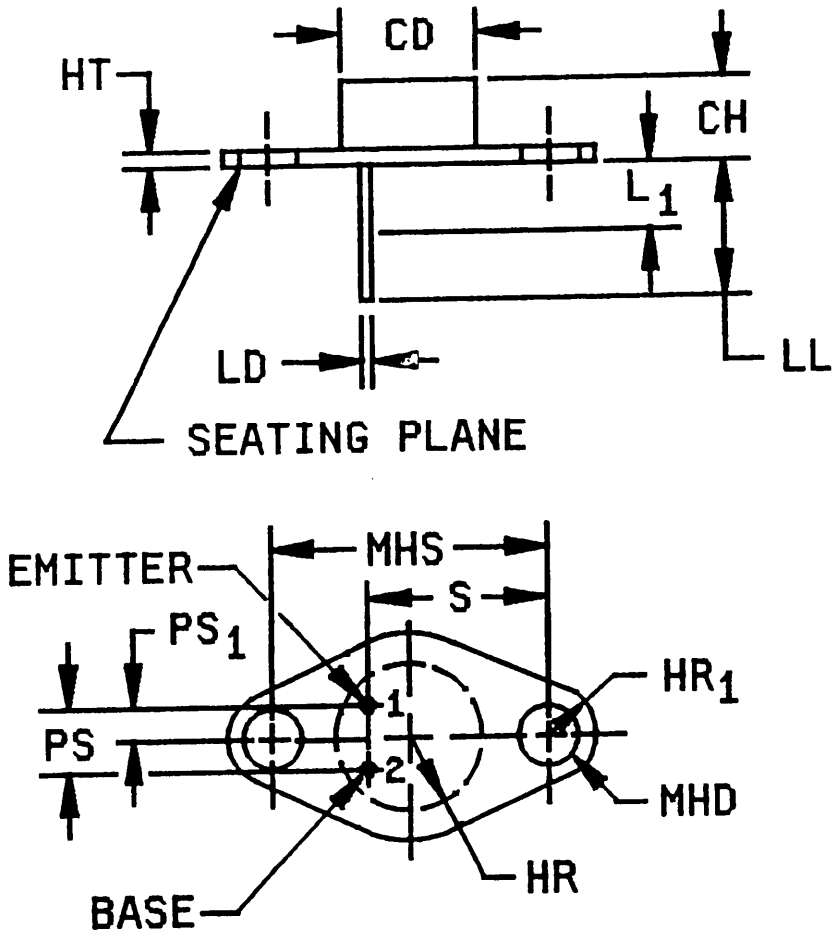


FIGURE 1. Physical dimensions (TO-66).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.250	.340	6.35	8.64	
LD	.028	.034	0.71	.86	3,7,10
CD	.470	.500	11.94	12.70	3,10
PS	.190	.210	4.83	5.33	4
PS_1	.093	.107	2.36	2.72	4
HT	.050	.075	1.27	1.91	
LL	.360	.500	9.14	12.70	3,9
L_1		.050		1.27	9
MHD	.142	.152	3.61	3.86	7,10
MHS	.958	.962	24.33	24.43	
HR		.350		8.89	
HR_1	.115	.145	2.92	3.68	6
s	.570	.590	14.48	14.99	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Body contour is optional within zone defined by CD and PS_1.
4. These dimensions should be measured at points .050 inch (1.27 mm) .055 inch (1.40 mm) below seating plane. When gauge is not used measurement will be made at the seating plane.
5. Both terminals.
6. At both ends.
7. Two holes.
8. The collector shall be electrically connected to the case.
9. LD applies between L_1 and LL. Diameter is uncontrolled in L_1.
10. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions - Continued.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels
3c 1/ 2/	Thermal impedance (see 4.5.3)
9	I_{CEX1}
11	h_{FE2}; I_{CEX1}
12	See 4.3.1
13	ΔI_{CEX1} = 100 percent of initial value or 1 mA dc, whichever is greater. Δh_{FE2} = ±25 percent of initial value; subgroup 2 of table I herein.

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.
2/ $Z_{\theta JX}$ is not required in screen 13 if already previously performed.

4.3.1 Power burn-in. Power burn-in conditions are as follows:

T_J = +187.5°C, ±12.5°C; $V_{CB} \geq$ +10 V dc; T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection: $Z_{\theta JX} \leq 35°C/W$.

a. I_M measurement current . . . . . 1 to 10 mA.

b. I_H forward heating current . . . . . .5 A to 1.0 A.

c. t_H heating time . . . . . 10 ms.

d. t_{MD} measurement delay time . . . . . 70 µs maximum.

End-point electrical measurements shall be in accordance with the applicable steps of table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500, and as follows herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table I, group A, subgroup 2 herein, except $Z_{\theta JX}$ shall be performed after intermittent operation life only.

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4.4.2.1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Conditions
3	1037	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum.
5	3131	See 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows herein. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table I, group A, subgroup 2 herein, except $Z_{\theta JX}$ shall be performed after intermittent operation life only.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
2	2036	Test condition A; weight = 10 lbs; t = 15 s.
6	1037	Test condition D, 6,000 cycles. The heating cycle shall be 1 minute minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC}$ (max) = 5°C/W.

a. Collector current magnitude during power applications shall be 0.55 A dc.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature point shall be the case.

d. Reference point temperature shall be +25°C ≤ T_R +25°C ≤ +75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

4.5.3 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750C, method 3161. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for engineering evaluation and disposition. One hundred percent Safe Operating Area (SOA) testing may be performed in lieu of thermal impedance testing herein provided that the appropriate conditions of temperature, time current, and voltage to achieve die attach integrity are submitted to the qualifying activity.

4.5.3.1 Thermal impedance $Z_{\theta JX}$ measurements for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750C, method 3161 (read and record data ($Z_{\theta JX}$). $Z_{\theta JX}$ shall be supplied on one lot (500 pieces minimum) and a thermal response curve shall be submitted. Twenty-two of these samples shall be serialized and provided to the qualifying activity for correlation prior to shipment of parts. Measurements conditions shall be in accordance with 4.4.1.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias condition C; I_C = 200 mA dc; Pulsed (see 4.5.1)	$V_{(BR)CEO}$		75	V dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 50 V dc	I_{CEO}		5	mA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 100 V dc; V_{BE} = 1.5 V dc	I_{CEX1}		4	mA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO}		10	mA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 120 V dc	I_{CBO}		25	mA dc
Base emitter voltage (nonsaturated)	3066	Test condition B; I_C = 4.0 A dc; V_{CE} = 2.0 V dc; Pulsed (see 4.5.1)	V_{BE}		1.8	V dc
Base emitter voltage (saturated)	3066	Test condition A; I_C = 4.0 A dc; I_B = 0.4 V dc; Pulsed (see 4.5.1)	$V_{BE(SAT)}$		2.0	V dc
Collector to emitter saturated voltage	3071	I_C = 4.0 A dc; I_B = 0.4 V dc; Pulsed (see 4.5.1)	$V_{CE(SAT)}$		1.2	V dc
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 0.5 A dc; Pulsed (see 4.5.1)	h_{FE1}	40		
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 4.0 A dc; Pulsed (see 4.5.1)	h_{FE2}	20	80	
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 4.0 A dc; Pulsed (see 4.5.1)	h_{FE3}	12	100	
Thermal impedance	3131	See 4.5.3	$Z_{\theta JX}$		5.0	°C/W

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 100 V dc; V_{BE} = 1.5 V dc	I_{CEX2}		4.0	mA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 0.5 A dc; Pulsed (see 4.5.1)	h_{FE4}	10		
Subgroup 4						
Pulse response	3251	Test condition A except test circuit and pulse requirements in accordance with figure 2 herein.				
Turn-on time		V_{CC} = 30 V dc; I_C = 4.0 A dc; I_B = 0.4 A dc;	$t_{(on)}$		0.44	µs
Turn-off time		V_{CC} = 30 V dc; I_C = 4.0 A dc; $I_B = -I_B$ = 0.4 A dc;	$t_{(off)}$		1.2	µs
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 500 mA dc; f = 10 MHz	$\|h_{fe}\|$	4	20	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 0.1 MHz ≤ f ≤ 1.0 MHz	C_{obo}		175	pF
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t = 1 s, 1 cycle; (See figure 3)				
Test 1		V_{CE} = 5 V dc; I_C = 7 A dc				
Test 2		V_{CE} = 28 V dc; I_C = 1.25 A dc				
Test 3		V_{CE} = 40 V dc; I_C = 500 mA dc				
Test 4		V_{CE} = 75 V dc; I_C = 100 mA dc				

See footnote at end of table.

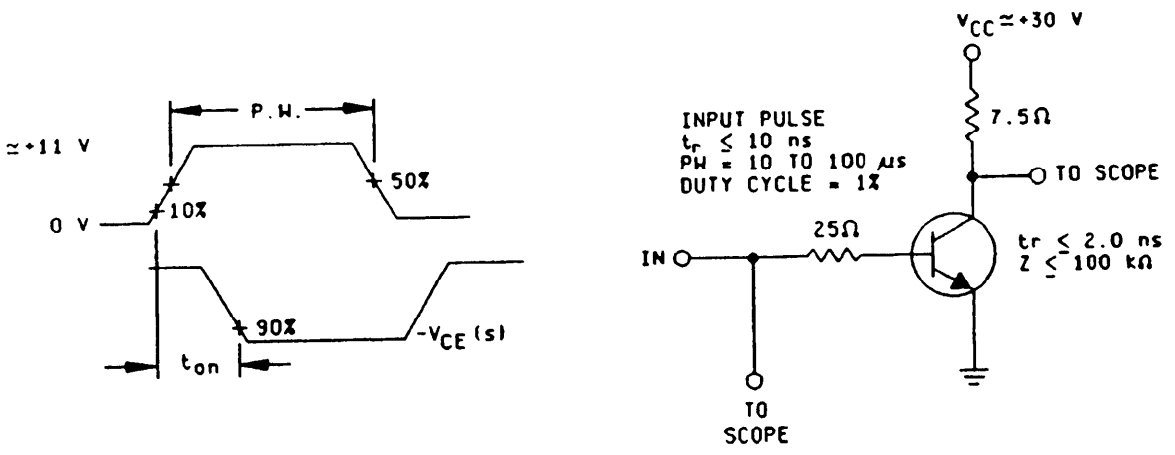
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TABLE I. Group A inspection - Continued.

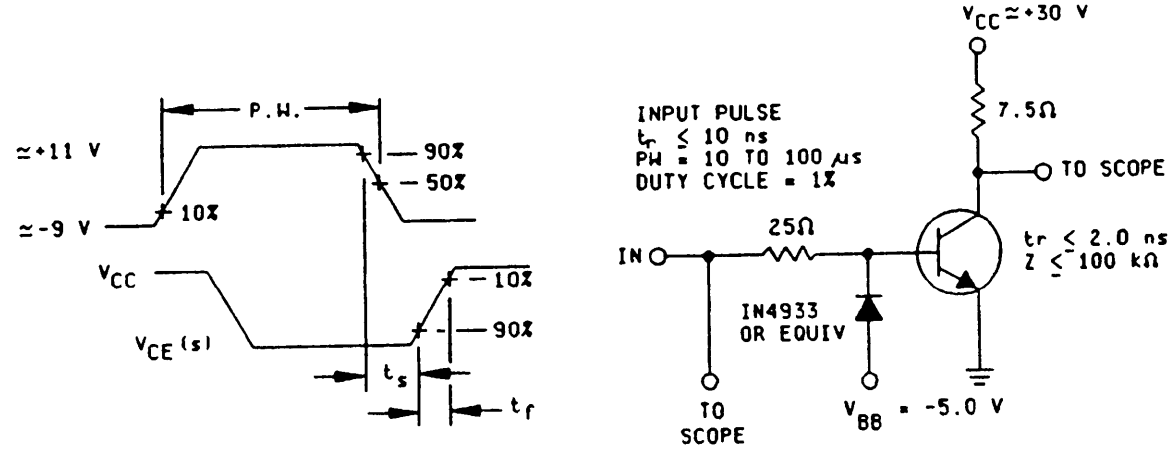
Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (switching)	3053	Load condition C, (unclamped inductive load) (see figure 4) T_C = +25°C; duty cycle ≤ 5 percent; R_S = 0.1 ohm; $t_r = t_f \leq 2$ µs				
Test 1		$t_p \approx 50$ µs (vary to obtain I_C); R_{BB1} = 5.0 ohms; V_{BB1} = 10 V dc; R_{BB2} = 50 ohms; V_{BB2} = 4 V dc; I_C = 4 A dc; L = 250 µH; V_{CC} = 10 V dc; R of inductor = 0.1 ohm				
Test 2		$t_p \approx 25$ µs (vary to obtain I_C); R_{BB1} = 5.0 ohms; V_{BB1} = 10 V dc; R_{BB2} = 50 ohms; V_{BB2} = 1.5 V dc; I_C = 1 A dc; L = 250 µH; V_{CC} = 10 V dc; R of inductor = 0.1 ohm				
Safe operating area (switching	3053	Load condition B; T_A = +25°C; I_C = 7 A dc; V_{CC} = 45 V dc; Clamped voltage = 75 V dc, duty cycle ≤ 5 percent; R_{BB1} = 5.0 ohms; V_{BB1} = 10 V dc; R_{BB2} = 50 ohms; V_{BB2} = -4 V dc; L = 250 µH; R of inductor = 0.1 ohm; R_L = as required to limit I_C.				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TURN-ON (t_{on}) TIME TEST CIRCUIT



TURN-OFF (t_{off}) TIME TEST CIRCUIT

FIGURE 2. Switching time test circuits.

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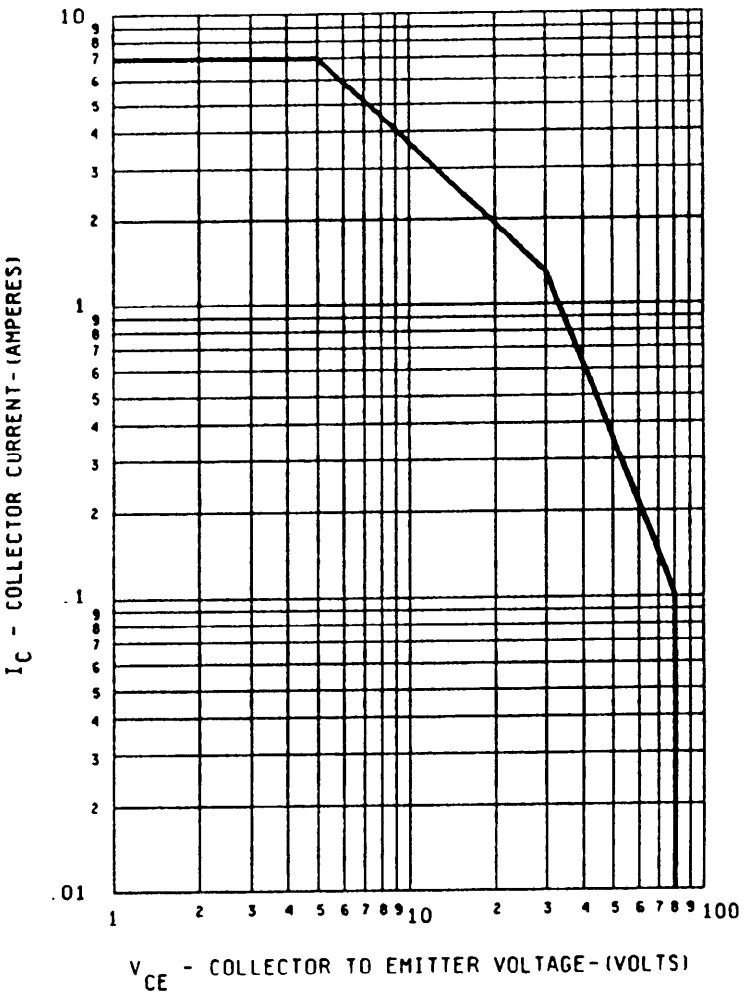


FIGURE 3. Maximum safe operating graph (continuous dc).

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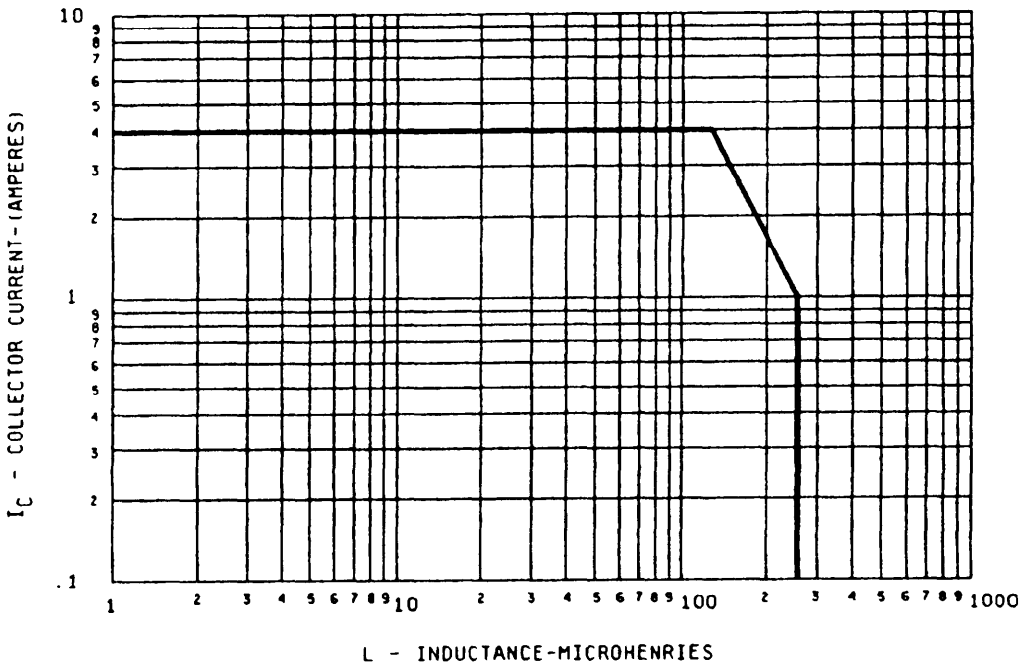


FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped load).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Air Force - 13, 15, 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1563)

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/526C	2. DOCUMENT DATE (YYMMDD) 950320

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER TYPES 2N3879 JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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